Sherif Morsi

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Summary

Concordia Computer Science student seeking AI/ML, software, and quant internships. Trained **PyTorch** models on 5G telemetry at Nokia; builds tool-calling **AI agents**, QLoRA fine-tunes, and leakage-safe forecasting models in tested **Python**.

Education

Concordia University Montreal, QC, Canada
Bachelor of Computer Science *Expected May 2028*

- **Availability:** Winter 2027 (Jan to Apr 2027) for 4, 8, 12, or 16 months
- **Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Artificial Intelligence, Machine Learning, Linear Algebra, Multivariable Calculus

Experience

AI & Machine Learning Intern Aug 2026 - Sep 2026
Nokia *Cairo, Egypt*

- Trained and evaluated Python and **PyTorch** models on wireless network telemetry to forecast 5G **energy demand**
- Developed **data and ML pipelines** for 5G energy telemetry, to support automated network management and optimization
- Presented findings on wireless networking and 5G-related energy challenges to cross-functional teams of data scientists and engineers; led the project proposal and requirements review

Credit Analyst Job Shadow Aug 2025 - Sep 2025
Qatar National Bank (QNB) *Cairo, Egypt*

- Shadowed credit analysts reviewing SME financial records as part of the bank's credit-review process
- Built **Python** utilities and **SQL** queries to preprocess and validate SME financial records, improving the data quality and reliability of credit-review data
- Tested and documented these scripts and fixes with the team so they stayed maintainable and easy to debug

Projects

Orbit (Agentic AI, Multi-Agent Tool Calling) | *Python, PyTorch, QLoRA, Claude and Gemini APIs* | **GitHub** 2026

- Built a planner, worker, and critic **multi-agent** system with JSON-schema **tool calling** over 6 sandboxed tools
- Added **guardrails**: a grounding gate refuses unsupported claims, and the critic forces up to 2 revisions
- Fine-tuned **Qwen2.5-1.5B** with **QLoRA**: schema-valid tool calls rose **0.61 to 0.78** on a frozen test (95% CI above zero)
- Built Claude API **distillation** with hard cost caps and a trace-replay dashboard; **94 pytest tests** in CI

Time-Series Foundation Model (Transformers, Forecasting) | *Python, PyTorch, pytest, GitHub Actions* | **GitHub** 2026

- Achieved **0.605**× **Seasonal Naive** zero-shot validation CRPS by pretraining a **10.1M-parameter** PyTorch **transformer**
- Validated **443,258 series** across **77 datasets** to keep evaluation benchmarks leakage-free, excluding **801** flagged series
- Implemented the transformer, training loop, and GIFT-Eval scoring in plain PyTorch; trained in fp16 on a single T4 GPU
- Scored a smaller model once on the GIFT-Eval test split (**97 configs, 3 seeds**); **150 automated tests** in GitHub Actions CI

World Shock Lab (Econometrics, Macro Shocks) | *Python, PyTorch, DuckDB, TypeScript* | **GitHub** | **Live Site** 2026

- Estimated Fed/BoC/OPEC surprise effects on **53 global assets**: a 10 bp Fed surprise moves the TSX −**1.10%**, BoC −**0.19%**
- Validated estimators by **simulation** and replicated **11 of 13** published findings; quantile boosting won on 5% tail loss
- Shipped a React/TypeScript scenario tool (under **3 s** on a throttled phone) whose tests fail on any number not in the results

Certifications & Affiliations

Anthropic: Building with the Claude API, Claude Code 101, Claude 101 | **IBM:** Machine Learning with Python
Quantitative Research and Competitions Club (QUARCC) | *Member, Concordia University*: Exploring machine learning concepts, statistical modeling, and quantitative research through data-driven problem-solving

Technical Skills

Languages: **Python**, **SQL**, C++, **Java**, C, R, JavaScript, TypeScript, Bash, MATLAB
Machine Learning: **PyTorch**, TensorFlow, **scikit-learn**, Deep Learning, Supervised/Unsupervised Learning, Regression, **Transformers**, **Time-Series Forecasting**
ML Methods: **Feature Engineering**, **Cross-Validation**, Transfer Learning, Computer Vision, NLP, **Model Evaluation**, Mixed Precision
AI / LLMs: **LLMs**, **AI Agents**, **Tool Calling**, **Prompt Engineering**, QLoRA, Distillation, Claude API, Hugging Face
Quant / Risk: **Econometrics**, Event Studies, Monte Carlo, Probabilistic Forecasting, **CVaR**, **Portfolio Optimization**, Walk-Forward Validation
Data / Libraries: **Pandas**, **NumPy**, SciPy, Matplotlib, OpenCV, **FastAPI**, **DuckDB**, **SQLite**, Parquet, cvxpy, statsmodels
Developer Tools: **Git**, **GitHub Actions (CI/CD)**, **pytest**, Playwright, Jupyter, Colab, Claude Code, Cursor, Codex, VS Code
Office / BI: Excel (Advanced), Power BI, PowerPoint, Microsoft 365